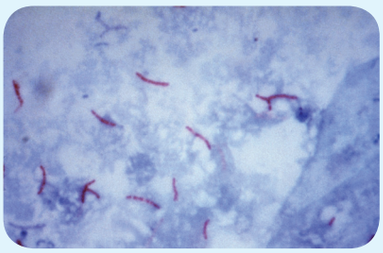
DO YOU KNOW?

Mycobacterium leprae which causes leprosy is an unculturable bacterium. It is primarily diagnosed by using a special bacteriological stain called Acid Fast stain.



Mycobacterium leprae (Acid Fast bacilli) stained with modified Ziehl Neelson stain.

Methylene blue is more frequently used than any other stain in Bacteriology. It is used for the rapid survey of bacterial population of milk. It is also used for the diagnosis of Diphtheria. This stain is incorporated along with Eosin in Lactose Agar to distinguish *Escherichia coli* from other fecal bacteria in contaminated water.

3.7 Differential Staining

In this method more than one stain is employed. In some method the stains are applied separately, while in other method they are mixed and applied in one application. These procedures show differences between the cells or parts of a cell and can be used for of identification. The two most important differential stains used by bacteriologists are Gram stain and Acid Fast stain. The differences between simple and differential staining are shown in Table 3.3.

3.7.1 Gram's Staining Method

The Gram's stain technique was developed by Danish Bacteriologist Hans Christian Gram in 1884. It is one of the most useful staining methods because it classifies bacteria into two large groups namely Gram positive and Gram negative. In this method, the fixed bacterial smear is subjected to staining reagents in the order of sequence listed below:

Crystal violet
(Primary stain)
↓
Iodine
(Mordant)
↓
Alcohol/Acetone
(Decolourising agent)
↓
Basic fuschsin/Safranin
(Counter stain)

Table 3.3: Differences between Simple and Differential Staining

Simple staining	Differential staining
1. This method uses only one stain.	This method uses more than one stain.
2. It imparts only one colour to all bacterial cells.	It imparts two or more different colours to bacterial cells.
3. It reveals the size, shape and arrangement of bacterial cells.	It reveals the size, shape and arrangement. In addition, it differentiates two groups of bacteria.
Example: Methylene blue staining method.	Example: 1. Gram's staining method 2. Acid Fast staining method

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